[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, BALL SHAPE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers ball shape, radius nose, high speed steel or tungsten carbide type rotary files intended for general-purpose machining and deburring.

2. CLASSIFICATION

2.1 Classification. The rotary files shall be of the following types, styles, sizes and classes. The type, style, size and class to be furnished shall be as specified (see 7.2).

Type: Type I - High speed steel.
Type II - Tungsten carbide.

Style: Style A - Right hand spiral fluting.
Style B - Right hand spiral fluting with chip breaker.
Style C - Double cut fluting.
Style D - Diamond cut fluting.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3455

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Size: Unless otherwise specified (see 7.2), the cutter head diameter and shank diameter of the type I and type II rotary files shall conform to tables I and II, respectively.

TABLE I. Type I, high speed steel rotary files, sizes, and characteristics.

Size	Characteristics	
	Shank Diameter (in.)	Cutter Head Diameter (in.)
1	1/8	1/4
2	1/8	3/8
3	1/4	1/8
4	1/4	3/16
5	1/4	1/4
6	1/4	5/16
7	1/4	3/8
8	1/4	1/2
9	1/4	5/8
10	1/4	3/4
11	1/4	1

TABLE II. Type II, tungsten carbide rotary files, sizes, and characteristics.

Size	Characteristics	
	Shank Diameter (inches)	Cutter Head Diameter (in.)
1	1/8	3/32
2	1/8	1/8
3	1/8	1/4
4	1/4	1/8
5	1/4	3/16
6	1/4	1/4
7	1/4	5/16
8	1/4	3/8
9	1/4	7/16
10	1/4	1/2
11	1/4	5/8
12	1/4	3/4
13	1/4	7/8
14	1/4	1

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Class: Unless otherwise specified (see 7.2), the number of teeth around the circumference of the cutter head shall be selected from tables III and IV for the diameter of cutter head specified.

TABLE III. Type I, rotary files, number of teeth per circumference.

Cutting Head Diameter (in.)	Teeth Per Circumference			
	Class A Coarse Cut	Class B Medium Cut	Class C Fine Cut	Class D Extra Coarse Cut
1/8	6 - 8	9 - 11	12 - 15	4 - 5
3/16	9 - 11	12 - 14	15 - 20	4 - 8
1/4	10 - 13	14 - 16	17 - 21	4 - 9
5/16	11 - 16	17 - 20	21 - 26	5- 10
3/8	12 - 17	18 - 22	23 - 30	5 - 11
1/2	14 - 20	21 - 26	27 - 35	7 - 13
5/8	17 - 22	23 - 29	30 - 40	9 - 16
3/4	18 - 24	25 - 32	33 - 44	10 - 17
1	20 - 27	28 - 41	42 - 55	14 - 19

TABLE IV. Type II, rotary files, number of teeth per circumference.

Cutting Head Diameter (in.)	Teeth Per Circumference			
	Class A Coarse Cut	Class B Medium Cut	Class C Fine Cut	Class D Extra Coarse Cut
1/16	7 - 10	11 - 13	14 - 16	4 - 6
3/32	8 - 11	12 - 14	15 - 17	4 - 7
1/8	8 - 11	12 - 15	16 - 20	4 - 7
3/16	10 - 12	13 - 16	17 - 24	4 - 9
1/4	12 - 15	16 - 20	21 - 26	4 - 11
5/16	13 - 16	17 - 22	23 - 30	5 - 12
3/8	13 - 18	19 - 26	27 - 38	5 - 12
7/16	15 - 20	21 - 27	28 - 36	6 - 14
1/2	16 - 22	23 - 28	29 - 39	7 - 15
5/8	18 - 25	26 - 38	39 - 47	9 - 17
3/4	19 - 28	29 - 38	39 - 49	10 - 18
7/8	22 - 29	30 - 37	38 - 55	14 - 21
1	24 - 32	33 - 43	44 - 62	15 - 23

3. SALIENT CHARACTERISTICS

3.1 General requirements. The cutting heads of type I rotary files shall meet the minimum requirements for standard high-speed steel described in the annex of ASTM A600 - 92a when tested in accordance with ASTM A370. The cutting heads of type II rotary files shall be

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manufactured from tungsten carbide with a hardness of not less than 91.2 and not greater than 92.5 on the Rockwell “A” scale when tested in accordance with section ASTM B294. The cutting heads of type II rotary files shall have a density of 14.6 to 15.1 grams per cubic centimeter when tested in accordance with ASTM B311. Type II rotary files shall have a transverse rupture strength of 220,000 pounds per square inch when tested in accordance with ASTM B406.

3.2 Fluting. The fluting of the rotary file shall be either right-hand spiral, right-hand spiral with chipbreaker, double cut, or diamond cut, depending upon the style specified. The style A rotary files shall have right-hand spiral fluting consisting of a right-hand helix. The helix angle measured from the centerline of the cutter head shall be between 15° and 35°. Style B rotary files shall have right-hand spiral fluting with a left-handed spiral chipbreaker along the periphery of the cutting teeth. Style C rotary files shall have double cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that result in a wedge or chisel form cutting tooth. The number of left-hand spirals shall be from 40 to 60 percent of the number of right-hand spirals. Style D rotary files shall have diamond cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that result in a pyramidal form cutting tooth. The number of left-hand spirals shall be from 70 to 90 percent of the number of right-hand spirals.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description; conform to the producer’s own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified (see 7.2).

7. NOTES

7.1 Source of documents.

7.1.1 Industry documents. Copies of industry standards referenced in this CID may be obtained from the following address:

American Society for Testing and Materials (ASTM)

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ASTM A370 -Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A600 - Standard Specification for Tool Steel High Speed

ASTM B294 -Standard Test Method for Hardness Testing of Cemented Carbides

ASTM B311 - Test Method for Density Determination for Powder Metallurgy (P/M) Materials Containing Less Than Two Percent Porosity

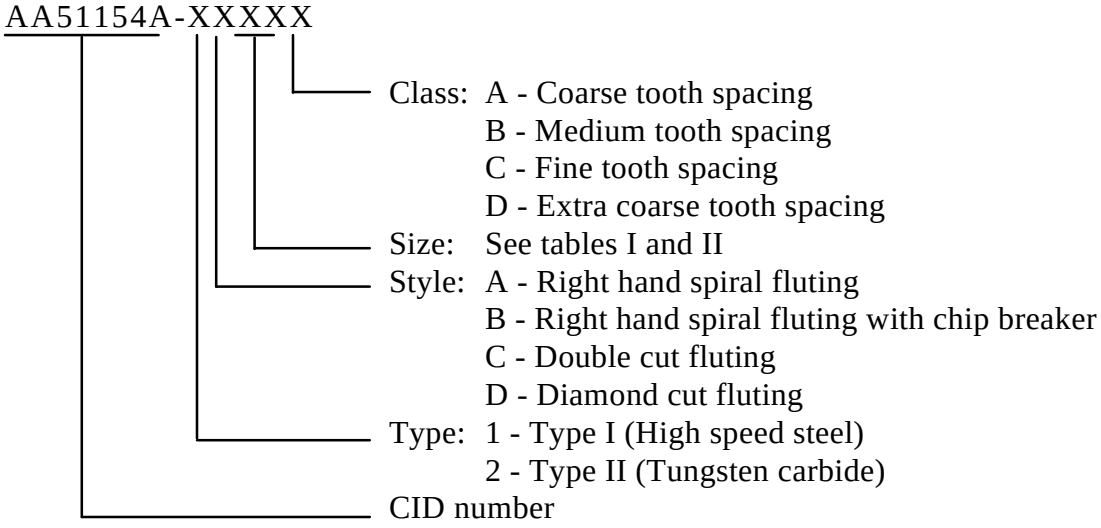
ASTM B406 -Standard Test Method for Transverse Rupture Strength of Cemented Carbides

Application for copies should be addressed to American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, style, size, and class as required (see 2.1).
c. Shank diameter, if different (see 2.1).
d. Cutter head diameter, if different (see 2.1).
e. Cutter head teeth per circumference (see 2.1).
f. Packaging requirements (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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(Project 3455-0156)

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